

# Lecture 3: jQuery

Course: Web Programming 2

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- Introduction to jQuery – A lightweight JavaScript library simplifying HTML manipulation, event handling, animations, and AJAX.

# Why Use jQuery?

- Shorter and easier syntax than JavaScript
- Cross-browser compatibility
- Built-in effects and animations
- Simplifies AJAX and DOM manipulation
- Example:
- JS:  
`document.getElementById('demo').style.display='none';`
- jQuery: `$('#demo').hide();`

# jQuery Syntax

- Basic Format: `$(selector).action();`
- Example:
- `$('#btn').click(function(){ alert('Button clicked!'); });`
- Explanation:
- `#btn` selects the element with id 'btn'
- `.click()` adds an event listener
- `alert()` displays a message.

# jQuery Selectors

- `#id` → `$('#title')`
  - `.class` → `$('.item')`
  - `element` → `$('p')`
  - `this` → `$(this)`
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- Explanation: Select elements like CSS and perform actions using jQuery.

# jQuery Events

- Definition: jQuery events are user actions such as clicks or keypresses.
- Common Events:
- `click()`, `dblclick()`, `mouseenter()`, `mouseleave()`, `keypress()`, `keyup()`, `keydown()`, `submit()`, `change()`
- Example:
- `$('#btn').click(function(){ $('p').toggle(); });`

# Multiple Events Example

- `$('#box').on({`
- `mouseenter: function(){`  
`$(this).css('background-color','lightblue'); },`
- `mouseleave: function(){`  
`$(this).css('background-color','white'); },`
- `click: function(){ $(this).css('background-`  
`color','green'); }`
- `});`

# Keyboard and Form Events

- Keyboard Example:
- `$('#input').keyup(function(){  
 $('#display').text($(this).val()); });`
- Form Example:
- `$('#myForm').submit(function(event){  
 alert('Form submitted!');  
 event.preventDefault(); });`



# jQuery Effects

- `.fadeIn()` – shows the element
- `.fadeOut()` – hides the element
- `.slideToggle()` – slides up/down
- Example:
- `$('#box').animate({left:'250px',opacity:'0.5'},1000);`

# HTML & CSS Manipulation

- `$('#title').text('Welcome!');`
- `$('#img').attr('src','new.png');`
- `$('#div').css('background-color','yellow');`
  
- Explanation:
- `.text()` changes text
- `.attr()` changes attributes
- `.css()` modifies styling

# AJAX Example

- `$('#loadBtn').click(function(){  
 $('#data').load('info.txt'); });`
- Explanation: Loads content from a file into an element when button is clicked.

# CSS vs JavaScript vs jQuery

- CSS: Controls style & design
  - JavaScript: Controls logic & interactivity
  - jQuery: Simplifies JavaScript
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- Example:
  - CSS: `p {color:red;}`
  - JS:  
`document.getElementById('p').style.color='red';`
  - jQuery: `$('p').css('color','red');`

# Libraries Comparison

- JavaScript Library: Prewritten JS functions
- jQuery: Simplifies DOM manipulation and AJAX
- Bootstrap: Front-end framework for responsive design
- Explanation: JavaScript is core, jQuery simplifies it, Bootstrap focuses on styling.

# Practice Examples

- 1. Toggle Paragraph
- `$('#btn').click(function(){ $('#p').toggle(); });`
- 2. Animate Box
- `$('#box').animate({left:'200px',opacity:'0.7'},1000);`
- 3. Fade on Hover
- `$('#text').hover(function(){  
 $(this).fadeOut(500).fadeIn(500); });`

# Homework

- • Modify `.animate()` to move box up/down.
- • Add button to change page background with `.css()`.
- • Use `.fadeIn()` / `.fadeOut()` to make paragraph appear/disappear.

# HTML and jQuery Setup

- `<script  
src='https://code.jquery.com/jquery-3.6.0.min.js'></script>`
- Imports the jQuery library from CDN.



# Document Ready Function

- `$(document).ready(function() {`
- `// jQuery code runs here`
- `});`
- Ensures code executes after the HTML document is fully loaded.

# Toggle Paragraph Example

- HTML:
- `<p id='para'>This paragraph will hide and show.</p>`
- `<button id='toggleBtn'>Toggle Paragraph</button>`
- jQuery:
- `$('#toggleBtn').click(function() {`
- `$('#para').toggle();`
- `});`

# Animate Box Example

- HTML:
- `<div id='box'></div>`
- `<button id='move'>Animate Box</button>`
- jQuery:
- `$('#move').click(function() {`
- `$('#box').animate({ left: '250px', opacity: '0.5' },`  
 `1000)`
- `.animate({ left: '0px', opacity: '1' }, 1000);`
- `});`

# Slide Panel Example

- HTML:
- `<button id='slideBtn'>Show / Hide Panel</button>`
- `<div id='panel'>This panel slides up and down.</div>`
- jQuery:
- `$('#slideBtn').click(function() {`
- `$('#panel').slideToggle('slow');`
- `});`

# Form Handling Example

- HTML:
- `<form id='myForm'>`
- `<input type='text' id='name'>`
- `<button type='submit'>Submit</button>`
- `</form>`
- `<p id='greet'></p>`
  
- jQuery:
- `$('#myForm').submit(function(event) {`
- `event.preventDefault();`
- `let name = $('#name').val();`
- `if (name.trim() === '') {`
- `$('#greet').text('Please enter your name.');`
- `} else {`
- `$('#greet').text('Hello ' + name + '!');`
- `}`
- `});`

- **name.trim()**
  - Removes any extra spaces before or after the value of name.
  - Example: " dr.Wafaa ".trim() → "dr.Wafaa"
- **Condition: if (name.trim() === '')**
  - Checks if the user didn't type anything (i.e., the name is empty or only spaces).
  - If true → it means the input is blank.
- **\$('#greet').text('Please enter your name.')**
  - Finds the HTML element with **id="greet"** and changes its text to:
  - Please enter your name.
- **else { \$('#greet').text('Hello ' + name + '!'); }**
  - If the user entered a name, it will display:
  - Hello [name]!
  - Example: if the name is dr.Wafaa, it shows Hello dr.Wafaa!

# AJAX Example

- HTML:
- `<button id='loadBtn'>Load Text Data</button>`
- `<div id='data'></div>`
- jQuery:
- `$('#loadBtn').click(function() {`
- `$('#data').load('https://www.w3.org/TR/PNG/iso_8859-1.txt');`
- `});`

# Hover Effect Example

- jQuery:
- `$('#para').hover(`
- `function() { $(this).css('color', 'red'); },`
- `function() { $(this).css('color', 'black'); }`
- `);`
- Changes paragraph color on hover.



# Execution Flow Summary

- 1. Browser loads HTML and jQuery.
- 2. `$(document).ready()` runs after page load.
- 3. Event handlers activate.
- 4. AJAX loads external data dynamically.

# Summary Table

- Feature | jQuery Function | Description
- Toggle Paragraph | .toggle() | Hide/Show element
- Animate Box | .animate() | Move and change properties
- Slide Panel | .slideToggle() | Slide effect
- Form Handling | .submit(), .val(), .text() | Handle input
- AJAX | .load() | Load external data
- Hover | .hover() | Change color on hover

# Conclusion

- jQuery simplifies interactive web design.
- Enables animation, effects, and dynamic content.
- Always use `$(document).ready()` for best practice.

# H.W

- Modify `.animate()` to move box up/down.
- Add button to change page background with `.css()`.

Write a **jQuery program** that:

- Takes the user's **name** from an input field.
- Checks if the input is **empty**.
- Displays:
  - “Please enter your name.” if no name is entered.
  - “Hello [name]!” if the user enters a name.